Final report – HABI: Green Home Score & Retrofit Recommendation Engine

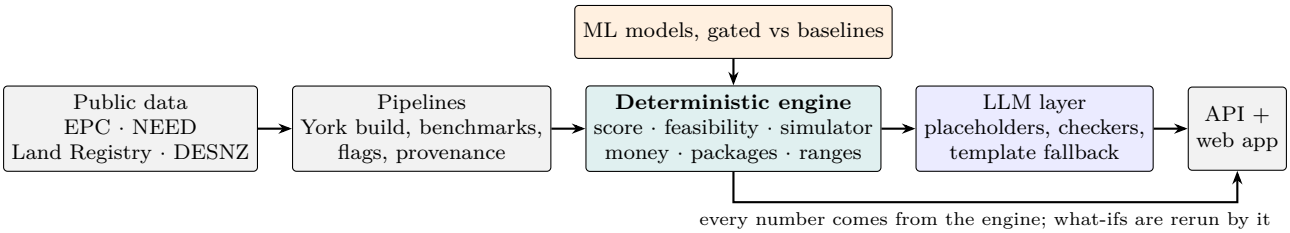
Megathon X: Singularity | Problem Statement 2 (Lloyds) | City of York

1 Solution

Every York home with an Energy Performance Certificate (**65,872 homes** from 95,348 certificates) gets a **Green Home Score out of 100** with a class, the retrofits it can *physically* take, and what each plan does to its **gas and electricity (reported separately), carbon, bills, EPC band and value**. Three design rules hold throughout:

- **One writer of numbers.** A deterministic engine (`scenario_builder.build`) computes every figure from public data with versioned configs; each result carries a reproducible scenario id (hash of inputs, request and versions) and the provenance of every value (observed, benchmark, modelled, assumption, user-provided, unavailable).
- **ML only where it beats a baseline.** LightGBM models add an indicative EPC change, a value range and uncertainty; each ships only if it beats a simple baseline on validation; the test split was scored once.
- **The LLM explains, it never calculates.** It writes sentences with placeholders that code fills from the engine result; what-if questions become validated engine requests and the engine is rerun.

A web app takes a homeowner from score to plan in four steps; a portfolio view ranks homes needing help for a lender or council. Missing data stays *missing* (null plus a status), never zero.



2 Approach

2.1 Data

EPC certificates and recommendations 2008–2026 (MHCLG, streamed from the 8.3 GB bulk file and filtered to York); DESNZ NEED anonymised data (4M sample) and NEED *Impact of measures* 2026 (measured savings of real installations); Land Registry Price Paid 2025–26; DESNZ 2026 carbon factors (gas 0.18231, electricity 0.14395 kg CO_2e/kWh); ECUK 2026 end-use split; Ofgem price cap Oct–Dec 2026; Boiler Upgrade Scheme. Homes are split train/val/test by home; score parameters, encoders and tuning use train only. Addresses and full postcodes are never loaded by models nor returned by the API.

Table	Rows	Notes
EPC certificates (York)	95,348	latest per home → 65,872 homes; split train 46,151 / val 9,797 / test 9,924
Assessor recommendations	231,548	on latest certificates; labels for the relevance ranking, indicative costs
Retrofit history (certificate pairs)	32,051	element changes between certificates; counterfactual checks of the SAP model
York sales linked to an EPC home	4,000	of 4,095 (3,958 exact, 42 fuzzy address matches; audited precision 1.00)
NEED sample	4,000,000	benchmarks by fuel × type × EPC band × floor area; Yorkshire 369,297
Score parameters	43,032	train homes with a usable baseline; intensity p_{10} 75.7, p_{90} 186.0 kWh/m^2/yr

Data-quality flags travel with each home instead of rows being dropped: 21,358 homes (32%) have a latest EPC older than 10 years; contradictory or missing heating and energy records are flagged; nine automated checks (unique ids, split isolation, no future EPC at sale, no zero-filled baselines, train-only parameters, provenance values, feature roles, licence-safe demo homes) run on every rebuild.

2.2 Green Home Score

$S = \sum_k w_k s_k$ over seven parts: EPC energy efficiency 25%, energy intensity 20%, insulation 15%, heating 15%, windows 10%, renewables 10%, smart controls 5%. Intensity is mapped between train-set percentiles. Classes: ≥80 Highly efficient, ≥65 Good, ≥50 Moderate, ≥30 **Poor**, else **Very poor** (flagged). A missing part gives a range instead of a guess; more than two missing parts makes the score unavailable. On the York build the mean score falls monotonically with the EPC band (A 79.6 → G 30.7); only 56 homes are unscored.

2.3 Retrofit engine

- **Feasibility** per measure (loft, cavity wall, solid wall, double glazing, smart thermostat, air-source heat pump, solar PV): feasible, infeasible (e.g. flat roof ⇒ no loft; already insulated) or needs verification; unknown is never treated as feasible.
- **Baseline** energy: NEED median of similar homes (fuel × type × EPC band × floor area, ≥30 homes per cell), replaced by the user's meter reading when given.
- **Simulator**: measures applied in physical order; gas split into cooking, space and water heating; savings from NEED measured impacts, shrunk by home segment (empirical Bayes); electric homes via the NEED electric-heating split; a

heat pump moves heat to electricity at a field-trial SCOP of 2.80 (2.44–3.5), so electricity *rises* and is shown as such. A NEED-calibrated figure is shown next to every combined package.
- **Money**: bills at the price cap, grants, net cost, simple payback and an *illustrative* loan (not a lender product or offer).
- **Prioritisation**: every combination of feasible measures (≤128) is evaluated, the Pareto front kept, and three plans featured: *cheapest cutting emissions by ≥10%*, *largest carbon cut*, *most carbon per pound*. Stability is tested by 13 one-at-a-time sensitivity reruns; a 300-draw Monte Carlo seeded by the scenario id gives P10–P90 ranges.

2.4 ML models (gated against baselines)

SAP regressor with monotone constraints (indicative EPC change); value model = gradient boosting + hedonic regression with a sector-clustered bootstrap and a split-conformal 80% interval; NEED energy quantile models (P10/P50/P90); per-measure classifiers ranking measures by how likely an EPC assessor is to recommend them (priority, while the install order stays physical). Optuna tuning, seed 42, deterministic LightGBM; every model version enters the scenario id, and a model is loaded only if its validation gate passed.

2.5 LLM layer

The engine result becomes a fact sheet with a stable key per value. The LLM sees keys and descriptions, **never values**, and writes `{placeholders}`. Code fills them; a rule checker rejects sentences with typed numbers, wrong directions (calling a rise a saving), infeasible advice, unsupported claims or missing caveats and replaces them with deterministic template sentences; an independent number checker matches every number in the final text to an engine value. What-ifs (measures, tariffs, grants, partial heat pump, own reading, budget) are parsed into the engine's validated request schema, with a keyword safety net for explicit requests; invalid or out-of-scope requests are refused with the reason. Without a provider, a template answers with the same numbers. Providers are tried in order (Groq, Gemini, others) behind a rate limiter and a disk cache.

3 Features

- **Homeowner app (four steps).** *Your home*: score, class, what lets it down, yearly gas, electricity and carbon labelled *benchmark* or *your reading*, an interactive house diagram. *Your options*: the three featured plans with cost after grants, bill change, payback, carbon, gas and electricity, and a spending limit that reruns the engine's package search. *Try changes*: add/remove measures, occupancy, heat-pump efficiency, tariff, own meter reading; the engine recalculates and a before/after diff is shown. *Your plan*: feasibility and permission notes, affordability, a downloadable decision record, a request letter and a plain-English report.
- **Expert view.** Score and sub-score contributions, per-fuel energy and carbon, bills/grants/payback/loan, P10–P90 ranges, indicative EPC and value range with caveats, all packages on a Pareto chart, a decision-stability matrix and a "Why this number?" drawer (inputs, provenance, method, assumptions, sources, versions) on every result.
- **Ask a question.** "How green is my home?", "What if I add a heat pump?", "What if gas costs 6p?", "My budget is £5,000" – grounded answers bound to the scenario id shown on screen.
- **Homes needing help.** Portfolio of Poor and Very poor homes ranked by improvement potential, fabric-only potential, carbon per pound or carbon saving; an opportunity view, not a credit decision.
- **Method & evidence.** Sources and licences, score formula, catalogue and feasibility rules, measured vs assumed savings, every model against its baseline, LLM evaluation. The app also runs fully offline on an engine snapshot of three demo homes.

Example (demo home A, 1967–75 detached gas house, EPC E). Score **44 (Poor) → 70 (Good)** with all feasible measures (loft and solid wall infeasible: flat roof, cavity walls). Gas −20,957 kWh/yr, electricity **+4,865 kWh/yr** (heat pump), carbon −3,120 kg CO_2e/yr (−69.6%); bill £3,023 → £2,633 a year; cost £16,537 after a £7,500 grant; payback 42 years; indicative EPC E → C; value change +2.9% to +7.2% (association, not a causal effect). The cheapest plan that cuts emissions by at least 10% is cavity-wall insulation alone (£1,000).

4 Results

4.1 ML models

York; home-level splits; tuning and gates on validation; test split scored once.

Model	Metric (test)	Model	Baseline(s)	n	Use
EPC SAP score	MAE (SAP points)	**3.46**	5.96 median / 4.48 ridge	9,919	EPC change
EPC band	within ±1 band	**99.8%**	96.3% median	9,919	
counterfactual pairs	MAE of ΔSAP, same home	**5.03**	6.66	880	
Property value	median abs. % error	**9.0%**	16.7% area median / 10.5% hedonic	619	value range
80% interval	coverage (gate 75–85%)	**82.1%**	–	619	gate **pass**
Energy (NEED 4M)	P10–P90 coverage	**0.80**	pinball 1,146 vs 1,166 cell quantiles	325k	ranges
Recommender	NDCG@3 over feasible measures	**0.995**	0.521 engine order / 0.981 prevalence	7,840	priority

- **SAP model:** R^2 0.80, exact band 72.5%; monotone in every "better state" input (on 10,000 homes no fabric, glazing, controls or PV upgrade lowers the modelled SAP); on held-out certificate pairs it tracks real changes (Pearson r 0.81). For boiler-to-heat-pump pairs the modelled change matches observed certificates (test +7.6 vs

+9.8 SAP points).

- **Value:** hedonic effect +0.19% of price per SAP point (sector-clustered bootstrap 0.11–0.27%, 99.6% of draws positive); robust to a spatial split (median error 10.8%) and a time split (9.9% vs 18.9% baseline).
- **Energy:** the NEED quantile models give household ranges whose P10–P90 interval covers 80% of held-out homes. As a point estimate, the engine keeps the NEED cell medians (models improved MAE by 1.5–1.6%, below the 2% switch-on threshold); NEED cannot be linked to individual York homes.
- **Recommender** (test PR-AUC vs prevalence): loft 0.90 vs 0.59, cavity 0.99 vs 0.59, glazing 0.99 vs 0.78, thermostat 0.51 vs 0.19, solid wall 0.99 vs 0.98, PV 1.00 vs 0.98; heat pumps use rules (too few labels). Agreement means "consistent with the EPC assessor", not "feasible".
- **Weak slices** (flagged at >1.5× overall error, shown in the model cards): SAP at EPC F/G (MAE 9–10), value for flats (15%).

4.2 Engine against measured data

Single measures use NEED measured savings directly. Combined fabric packages are checked against NEED's measured combinations (% of whole-home gas, tolerance ±5 pp):

Combination	NEED n	NEED median	Engine	Gap (pp)
Cavity wall + loft	2,297	8.4%	14.8%	+6.4
Solid wall + loft	1,609	14.3%	19.9%	+5.6
Solid wall + heating controls	2,192	9.2%	21.6%	+12.4
Solid wall + loft + heating controls	4,841	10.3%	24.0%	+13.8

The engineering model is optimistic for stacked measures, so every combined package is shown *with* its NEED-calibrated figure and a warning; one of the 13 sensitivity runs re-picks the plans on the calibrated figures. Hand-calculated demo homes are reproduced to 10^{-6}.

4.3 LLM

53 held-out questions (the brief's four questions, what-ifs, missing and conflicting data, infeasible requests, electricity rising, value not estimated, out of scope, prompt injection), run end to end through the API on the real engine:

Run	Grounded	Factual acc.	Unsupported	What-if success	All checks
Target	≥90%	≥90%	<5%	–	–
Real LLM (free-tier models)	**100%** (401)	**100%** (198)	**0%** (720)	100% (17)	53/53
Template explainer (no LLM)	100% (585)	100% (263)	0% (740)	100% (17)	53/53

The checker rejected 183 of 979 LLM-written sentences (acceptance 81.3%); rejected sentences never reach the user. 42 of 53 answers were LLM-written; 11 were template answers, 9 of them by design (declines, refusals, blocked measures). Refusal and route correctness were 100%, including the injection item.

5 Validation, safeguards and fixes

- **Leakage controls.** Splits by home; parameters, encoders and tuning on train only; value features from the certificate lodged *before* the sale; the test split scored by a script that refuses to run twice.
- **Tests.** Backend **757 passed** on the real data (engine to hand-calculated fixtures, reconciliation, monotonicity of every sub-score, feature contract, privacy guard, ML services, LLM checker, API); frontend 73 passed (the browser validates every response against the engine contract); data-integrity and leakage checks 9/9. A full scenario with packages, sensitivity and Monte Carlo takes under half a second.
- **Issues found on the real data and fixed.** Electrically heated homes (12.9% of York) had no fabric savings; the NEED electric-heating split now covers them (homes with measures but no estimate in a test sample: 18% → 4%). The after-retrofit state of a heat pump now follows York heat-pump certificates (electric homes: +12.4 SAP, none negative). The LLM's misreadings of explicit requests ("add a heat pump", a stated budget) are caught by a keyword safety net; patched answers no longer splice phrases, keep each caveat once and always cover payback.
- **Responsible AI.** Every number carries its provenance; benchmarks are labelled until a meter reading replaces them; infeasible measures are never recommended; value is reported as an association with caveats; the loan is illustrative only; nothing is a lending, eligibility or credit decision. Every design choice is recorded in a numbered decision log.

6 Limitations and future improvements

- Baselines are population benchmarks unless a reading is entered; linking smart-meter data would make them the home's own. 32% of EPCs are over 10 years old, so recorded features may be out of date (flagged).
- Oil and other non-gas, non-electric homes get no savings estimate rather than an invented one (no metered NEED baseline for those fuels).
- The value range rests on 20 months of York sales; repeat sales after retrofits would allow a causal estimate.
- Future improvements: calibrated combination factors as the default, customer-provided data replacing stale EPC fields, authentication and deployment for a pilot, and a larger live-LLM evaluation with an LLM-judge sample.